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Studierichting: Informatica
Oefeningenreeks 2B nr. 5.1

Oefening: $\mathrm{Bgsin}(x) + \mathrm{Bgcos}(x)$

Stel $x = \sin(\alpha)$ met $\alpha \in \left[0, \frac{\pi}{2}\right]$

Dan geldt $x = \sin(\alpha) = \cos\left(\frac{\pi}{2} - \alpha\right)$

Dus $\mathrm{Bgcos}(x) = \frac{\pi}{2} - \alpha$

$\Rightarrow \mathrm{Bgsin}(x) + \mathrm{Bgcos}(x) = \alpha + \left(\frac{\pi}{2} - \alpha\right) = \frac{\pi}{2}$

Conclusie:

$$\forall x \geq 0,\ x \leq 1 : \mathrm{Bgsin}(x) + \mathrm{Bgcos}(x) = \frac{\pi}{2}$$

References